

AgriData Explorer: Understanding Indian agriculture with EDA

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Key Problem Statement :

The core problem is the difficulty stakeholders face in accessing, analyzing, and making informed decisions based on fragmented, complex, and poorly managed agricultural data in India.

This results in:

- **Fragmented Data:** Agricultural data is scattered across different states, districts, and sources, making a unified view impossible.
- **Lack of Easy Access:** Farmers, policymakers, and researchers struggle to retrieve the necessary data efficiently.
- **Complexity & Difficult Analysis:** The sheer volume and complexity of the data make it hard for users to derive meaningful insights, identify trends, and understand regional disparities.
- **Suboptimal Decision-Making:** The lack of accessible, analyzed data hinders farmers' ability to optimize crop choices and prevents policymakers from effectively targeting areas for intervention.

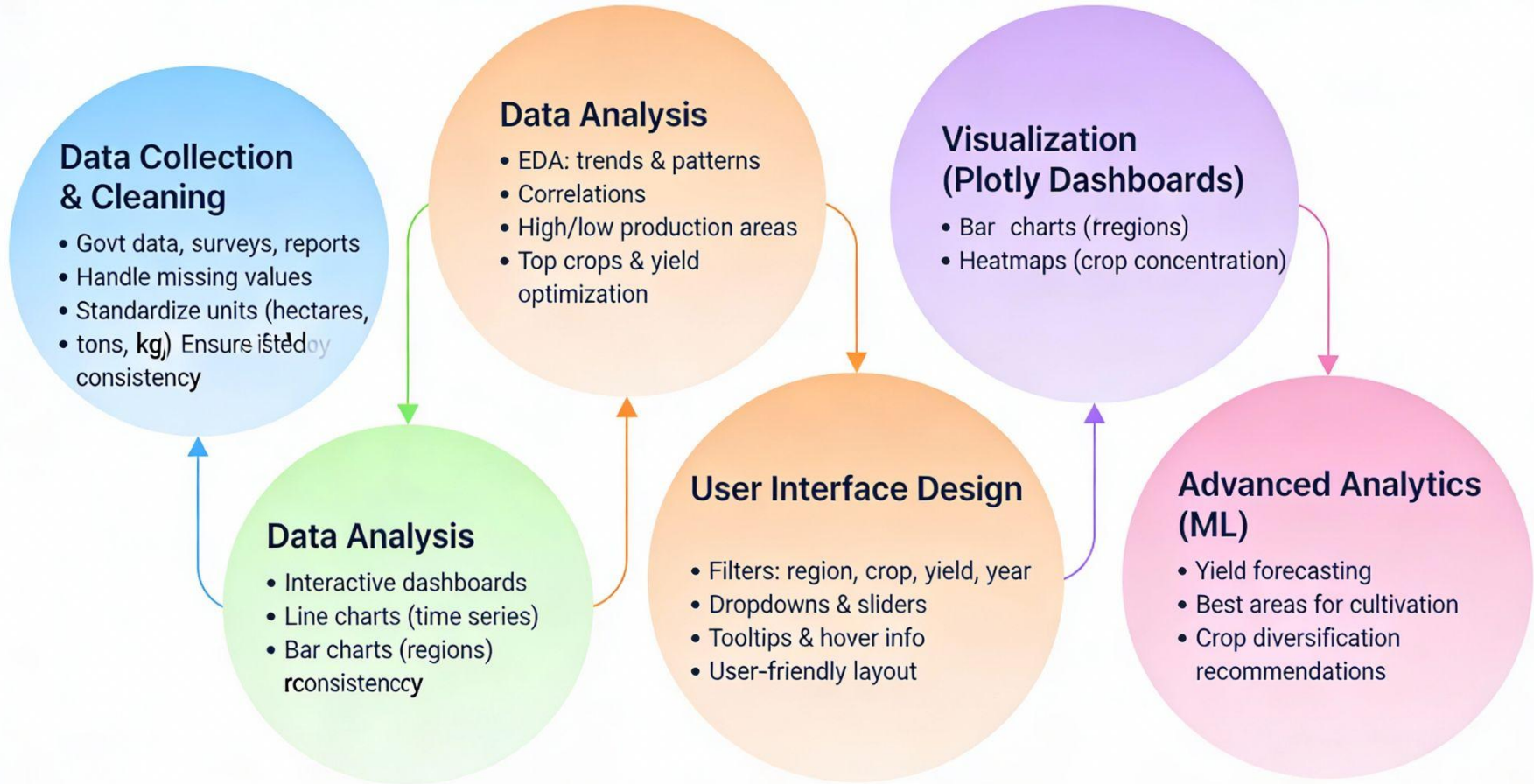


Project Goal and Solution Overview

The overarching goal is to **integrate agricultural data from various Indian states and districts into a single, user-friendly data visualization platform** to unlock actionable insights.

Key Solution Points:

- **Data Integration:**
 - Integrate agricultural data from **different states and districts** into a centralized system.
 - Create a **unified and comprehensive dataset** for India's agricultural sector.
- **Core Data Focus:**
 - Provide insights into key metrics such as **crop production, yields, and areas under cultivation**.
- **Visualization & Accessibility:**
 - Develop a **data visualization platform** that makes the complex data easy to access and understand.
 - Enable users to **identify trends, gaps, and regional disparities** quickly.
- **Stakeholder Benefits:**
 - **Farmers:** Optimize **crop choices** and farming practices based on regional data.
 - **Policymakers:** Assist in **targeting areas for intervention** (e.g., resource allocation, subsidy programs).
 - **Researchers:** Support in **analyzing agricultural trends** and understanding macro-level changes.



Power BI Agricultural Dashboard Development

Prepare Data

- Source data (CSV, Excel, SQL)
- Clear headers & formats
- Data fields (Crop, Region, Area, Production, Yield, Year)
- Multiple tables if needed

Interactive Filters

- Geographical Heatmaps
- Trend Lines & Bar Charts
- Scatter Plots
- KPI Indicators
- Cards & Gauges

Load Data into Power BI

- Open Power BI Desktop
- Get Data
- Choose data source
- Clean with Power Query

Create Data Model

- Define relationships
- Model View
- Establish table links
- Set Primary Keys

Design Dashboard

- Crop Type slicer
- Region slicer (state/district)
- Year slicer
- Drill-down capability

Enhance Visualizations

- Formatting (colors, labels)
- Interactivity between visuals
- Dynamic titles
- Tooltips

THANK YOU!!!